WEEK 3 OF 4

Particles

Ten thousand moving points, and how to get them on screen

No hand tracking today. That is deliberate.

Today's ideas are enough on their own. We connect them to your hands tomorrow.

Today is a detour

You have images and you have numbers. Today: things in 3D space.

Days 1 and 2 gave you

- TOPs — chains that change a picture
- CHOPs — streams of numbers from your hand
- Named Nulls, and "address, not a wire"

Today adds

- SOPs — actual geometry, points in 3D space
- A simulation that keeps its own state frame to frame
- A render chain — how anything 3D reaches your screen

Fair warning

This is the hardest week of the course — not because the clicking is difficult, but because more things can fail silently than on any other day.

There is a whole slide later listing the ways your screen can stay black with no error message. Expect to use it.

How this particle system works

Start with a grid of points. Push them around with a force. Draw what results.

1. **A grid gives you the points**

   Ten thousand of them, sitting in a flat sheet. They are not particles yet — just positions doing nothing.

2. **The Particle SOP makes them move**

   It takes those points as a starting arrangement and gives them physics: velocity, life, the ability to be pushed.

3. **A metaball is the thing that pushes**

   An invisible ball of influence. Move it through the points and they scatter away from it. This is the bit you will drive with your hand tomorrow.

A metaball is normally used to make blobby 3D shapes. Here we never draw it — we only use its field of influence as a force. It stays invisible all week.

Build it: the particles

Node names are what you type after pressing Tab.

1 `grid`

Set Rows and Columns to give you a few thousand points. Turn its viewer on — a flat mesh of dots.

2 `transform`

Wire the grid in. Use this to position and size the sheet. Handy later for framing.

3 `particle`

Wire Transform into its FIRST input. Turn its viewer on. The points should now drift — that is physics running.

4 `metaball`

On its own, not wired to anything yet. This is the invisible pusher.

5 `force`

Wire the metaball into it. This converts the metaball's field into a force particles can feel.

6 Connect the force

Wire Force into the Particle SOP's SECOND input. Move the metaball and the particles should scatter around it.

Making it visible

Geometry does not appear on screen by itself. It needs a container, a look, and an eye.

1 geo

A Geometry COMP. Go INSIDE it and delete whatever is already in there — it ships with a default shape that will hide your work.

2 in and out

Inside the geo, add an In SOP and an Out SOP and wire In into Out. These are the component's doorways: data enters through In, leaves through Out.

3 Wire the particles in

Back outside, wire particle1 into geo2's input. Your points are now flowing through the box.

4 constant

A Constant MAT. Drag it onto geo2's Material parameter. Flat colour, ignores lighting — which means you do not need a light at all.

5 camera

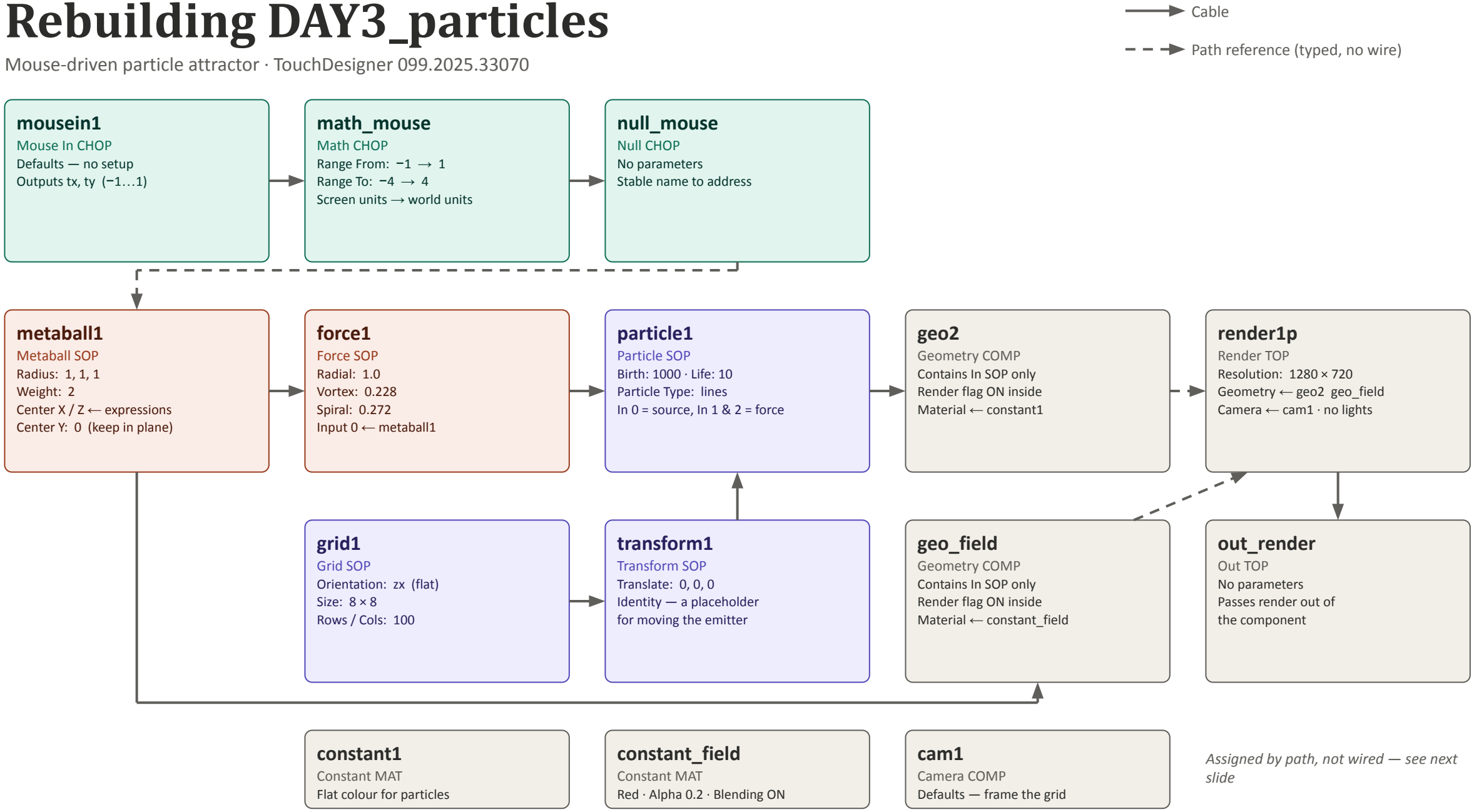
A Camera COMP. Pull its Translate Z back so it can see the scene.

6 render

A Render TOP. DRAG your camera and your geo onto its Camera and Geometry fields. Turn its viewer on.

Rebuilding DAY3_particles

Mouse-driven particle attractor · TouchDesigner 099.2025.33070



The links with no wires

Every connection above lives in a parameter field. Nothing in the network editor shows it.

Where	Parameter	What you type	Field colour
metaball1	Center X	`op('null_mouse')['tx']`	Blue
metaball1	Center Z	`op('null_mouse')['ty']`	Blue
metaball1	Center Y	`0  (leave constant)`	None
geo2	Material	`constant1`	None
geo_field	Material	`constant_field`	None
render1p	Geometry	`geo2  geo_field`	None
render1p	Camera	`cam1`	None

1. grid1 → transform1 → particle1. Nothing moves yet: no forces, no material.
2. metaball1 → force1, then force1 into particle1 inputs 1 and 2. Particles collapse to the origin.
3. geo2 with an In SOP, render flag on inside AND on the COMP. Material constant1. Drag geo2 and cam1 into render1p.
4. mousein1 → math_mouse → null_mouse. Watch the numbers; nothing on screen changes.
5. Type the two expressions into metaball1 Center X and Z. This is the moment the mouse takes hold.
6. geo_field + constant_field at alpha 0.2, added to render1p Geometry — the force field becomes visible.

Three mechanisms, one look

Expression — field turns blue. Evaluates live.
CHOP export — field turns green. Needs the source CHOP's export flag ON.
Export table — a Table DAT listing channel, target op and parameter.
An unfinished export leaves the field green and reading zero.

Two silent failures to stage

Render flag
Required on the SOP inside the Geometry COMP and on the COMP itself. One without the other renders nothing, with no warning.

Radius is reach
A metaball's field is exactly zero outside its radius. Shrink the blob and the force disappears — the patch looks broken but nothing is wrong.

Nothing is appearing

All of these produce an identical black screen with NO error message. Work down the list.

The Out SOP's render flag is off

Inside your geo, the Out node has a small square at its BOTTOM-RIGHT. That is the render flag, and new nodes do not inherit it. Bottom-left is 'let me see it', bottom-right is 'include it when rendering'. Different things.

The geo still has its default contents

A Geometry COMP arrives with a shape already inside. If you did not delete it, that shape renders and hides everything else.

The Render TOP's fields are empty

Camera and Geometry must actually be filled. DRAG the nodes in — typing a relative path can look accepted and silently stay empty.

The camera is pointing at nothing

It starts at the origin looking one way. Your particles may be behind it. Pull Translate Z back.

Your points are microscopic

At a small scale in a big scene they can be under a pixel. Temporarily scale everything up to check it exists at all.

You used a lit material

Phong and similar need a Light COMP in the scene or they render black. Constant MAT does not — that is why we use it.

A black render is not an error. TouchDesigner will not warn you, because an empty or unlit scene is a legitimate thing to want. Silence means it did exactly what you asked.

Driving it with the mouse

A rehearsal for tomorrow, using something simpler than a hand.

What to build

- Add a Mouse In CHOP. It gives you tx and ty straight away.
- Add a Math CHOP to rescale, then a Null called null_mouse.
- Drag null_mouse onto the metaball's Translate X, then again onto Translate Y.

Why this matters

Next week you swap null_mouse for null_tip_h1 and the whole thing becomes hand-controlled. One connection changes. Nothing else in the patch has to know.

That is exactly what the named Nulls were for.

If the metaball moves but the particles do not react

Check the metaball is actually reaching the particles — Force SOP wired into the Particle SOP's SECOND input — and that the metaball's radius is big enough to overlap the points. A tiny metaball in a large grid touches nothing.

If your machine is struggling

One number, and everyone stays on the same file.

Lower the grid's Rows and Columns. That is your particle count.

How to tell you need it

- The frame rate readout at the bottom of the window drops well below 60
- Your fan gets loud and the interface is lagging
- Dragging a parameter takes a moment to respond

What to do

Halve the rows and columns, and halve again if needed. Fewer particles looks less dense and behaves identically. Your file stays compatible with everyone else's, and tomorrow works the same.

This is not a lesser version of the exercise.

Everyone is running the same file with a different number in one field. Working out what your machine can handle is a normal part of this craft, not an admission of anything.

Homework: find one you like

Twenty minutes of play, one recording.

> *Tune it until it does something you actually find beautiful, then record ten seconds.*

1

Push it too far first

Take the metaball radius absurdly large, then tiny. Change the grid density. Find the edges before you look for the middle.

2

Then settle on one

Somewhere between those extremes is a setting you like. Trust that reaction — it is the only reliable instrument you have.

3

Write down what you changed

One or two lines with your recording. Naming what you did turns an accident into a technique.

Whatever you land on tonight is what your hands will control tomorrow.

So pick something you actually want to play with. Ten seconds of screen recording, as yesterday.

next week

WEEK 4 — Putting it together

Your hands take over from the mouse. Then we spend the rest of the hour making it look like something: different shapes, more forces, and post-processing on the render.

Before then

- [ ] Save your file. Particles and hand tracking both live in it now.
- [ ] Post your ten-second recording and what you changed.
- [ ] Check your MediaPipe still tracks — we go back to it tomorrow.
- [ ] make sure to close anything using your camera before you start!

Hardest day is behind you. Next week is the fun one.